

Unit 1 history of computing

Evolution of Computing and Information Technology

Evolution of computing means the gradual development of computers from simple calculating devices to powerful digital systems.

Main Stages

1. **Mechanical Era (Before 1940)**
 - Abacus → first counting device
 - Mechanical calculators (Pascal, Leibniz)
2. **Electromechanical Era**
 - Punch card machines
 - Early automatic computation
3. **Electronic Era (Generations of Computers)**
 - **1st Generation:** Vacuum tubes
 - **2nd Generation:** Transistors
 - **3rd Generation:** Integrated Circuits
 - **4th Generation:** Microprocessors
 - **5th Generation:** AI & Parallel Processing

Evolution of Information Technology (IT)

- Data processing → Information processing
- Standalone systems → Networked systems
- Manual records → Digital databases
- Local computing → Cloud computing

Q2

Historical Development of Computers

Early Devices

- Abacus → basic counting
- Napier's Bones → multiplication tool
- Pascaline → first mechanical calculator
- Analytical Engine → concept of modern computer

Modern Development

- Electronic computers in 1940s
- Stored program concept
- Commercial computers
- Microprocessor revolution
- Personal computers and Internet era

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Historical Development of Personal Computer (PC)

Early Phase (1970s)

Microprocessor invented

First hobby computers

Growth Phase (1980s)

Affordable home computers

Graphical interfaces introduced

Business adoption increased

Modern PC Era

High-speed processors

Laptops and portable devices

Multimedia and networking support

Integration with Internet and cloud

Q4Internet (Detailed Explanation)

Internet = Global network of interconnected computers that communicate using TCP/IP protocol.

Key Features

Worldwide connectivity

Information sharing

Communication platform

Resource sharing

Major Services

Email

File transfer

Online communication

E-commerce

Cloud services

How It Works

Devices connect via ISP

Data broken into packets

Q5WWW Full Name and Meaning

WWW = World Wide Web

It is a collection of web pages accessed through Internet using browsers.

- Uses HTTP protocol
- Contains websites, multimedia, hyperlinks

Q6 Emergence of Social and Ethical Problems in Computing

With technology growth, several issues emerged:

Social Problems

Privacy invasion

Cybercrime

Digital divide

Job displacement due to automation

Online addiction

Ethical Problems

Software piracy

Data misuse

Q7 Need of National Cyberspace Security

Cyberspace security protects national digital infrastructure.

Reasons

- Protection from cyber attacks
- Defense of banking and financial systems
- Safeguarding government data
- Protection of citizens' privacy

Q8Need of Computer Ethics Education

Computer ethics education teaches responsible technology use.

Importance

- Promotes legal software use
- Protects privacy and data
- Prevents cybercrime
- Develops professional responsibility